

Week 3 Open Problem

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1 Finite-Time Error in Theorem 1

Recall Theorem 1. For $\varepsilon > 0$, define the diffusively rescaled process

$$Q_t^\varepsilon = \varepsilon Q_{t/\varepsilon^2}.$$

Theorem 1 says that, as $\varepsilon \rightarrow 0$,

$$Q_t^\varepsilon \implies \text{Brownian motion with covariance matrix } D.$$

Equivalently, if we set

$$T = \frac{t}{\varepsilon^2},$$

then

$$\varepsilon = \sqrt{\frac{t}{T}},$$

and hence

$$Q_t^\varepsilon = \sqrt{\frac{t}{T}} Q_T.$$

In particular, if we fix $t = 1$, then the theorem suggests that

$$\frac{Q_T - Q_0}{\sqrt{T}} \implies N(0, D) \quad \text{as } T \rightarrow \infty.$$

The question in Week 3 Problem 3 is not only whether this convergence happens, but how fast it happens. Thus we want to understand how the distance

$$d\left(\mathcal{L}\left(\frac{Q_T - Q_0}{\sqrt{T}}\right), N(0, D)\right)$$

changes with T .

Since $Q_T - Q_0 \in \mathbb{R}^{dN}$, we study one-dimensional projections. Let

$$F \in \mathbb{R}^{dN}$$

be fixed, and define

$$Y_T^F := \frac{F^T(Q_T - Q_0)}{\sqrt{T}}.$$

Then Theorem 1 gives

$$Y_T^F \Longrightarrow N(0, \sigma_F^2),$$

where

$$\sigma_F^2 = F^T D F.$$

Therefore the finite-time question becomes:

How does $d(\mathcal{L}(Y_T^F), N(0, \sigma_F^2))$ change with T ?

In the numerical part, we use Kolmogorov distance:

$$d_K(X, Y) := \sup_{x \in \mathbb{R}} |\mathbb{P}(X \leq x) - \mathbb{P}(Y \leq x)|.$$

So in this problem,

$$d_K(\mathcal{L}(Y_T^F), N(0, \sigma_F^2)) = \sup_{x \in \mathbb{R}} |\mathbb{P}(Y_T^F \leq x) - \Phi_{\sigma_F}(x)|,$$

where Φ_{σ_F} is the distribution function of $N(0, \sigma_F^2)$.

1.1 Martingale-corrector decomposition

We now recall the key decomposition used in the proof of Theorem 1.

For simplicity, take $\beta = 1$. The wrapped process q_t satisfies

$$dq_t = -\nabla V(q_t) dt + \sqrt{2} dW_t.$$

The unwrapped process Q_t satisfies

$$Q_T - Q_0 = -\int_0^T \nabla V(q_s) ds + \sqrt{2} W_T.$$

Let Φ_0 solve the Poisson equation

$$\mathcal{L}\Phi_0 = -\nabla V,$$

where

$$\mathcal{L} = -\nabla V \cdot \nabla + \Delta$$

is the generator of the diffusion q_t .

By Itô's formula,

$$d\Phi_0(q_t) = \mathcal{L}\Phi_0(q_t) dt + \sqrt{2} \nabla \Phi_0(q_t) dW_t.$$

Since $\mathcal{L}\Phi_0 = -\nabla V$, we get

$$d\Phi_0(q_t) = -\nabla V(q_t) dt + \sqrt{2} \nabla \Phi_0(q_t) dW_t.$$

Integrating from 0 to T ,

$$\Phi_0(q_T) - \Phi_0(q_0) = -\int_0^T \nabla V(q_s) ds + \sqrt{2} \int_0^T \nabla \Phi_0(q_s) dW_s.$$

Rearranging,

$$-\int_0^T \nabla V(q_s) ds = \Phi_0(q_T) - \Phi_0(q_0) - \sqrt{2} \int_0^T \nabla \Phi_0(q_s) dW_s.$$

Substituting this into the formula for $Q_T - Q_0$, we obtain

$$Q_T - Q_0 = \Phi_0(q_T) - \Phi_0(q_0) + \sqrt{2} \int_0^T (I_d - \nabla \Phi_0(q_s)) dW_s.$$

Define

$$M_T := \sqrt{2} \int_0^T (I_d - \nabla \Phi_0(q_s)) dW_s.$$

By the theorem that stochastic integrals with square-integrable adapted integrands are martingales, M_T is a martingale. Hence

$$Q_T - Q_0 = \Phi_0(q_T) - \Phi_0(q_0) + M_T.$$

Plugging this into Y_T^F , we get

$$Y_T^F = \frac{F^T(Q_T - Q_0)}{\sqrt{T}} = \frac{F^T(\Phi_0(q_T) - \Phi_0(q_0))}{\sqrt{T}} + \frac{F^T M_T}{\sqrt{T}}.$$

We call

$$R_T^F := \frac{F^T(\Phi_0(q_T) - \Phi_0(q_0))}{\sqrt{T}}$$

the corrector part, and

$$Z_T^F := \frac{F^T M_T}{\sqrt{T}}$$

the martingale part. Therefore

$$Y_T^F = R_T^F + Z_T^F.$$

1.2 Part 1: Corrector part error

Since q_t lives on the torus and Φ_0 is smooth, Φ_0 is bounded. Thus there exists a constant $C_\Phi > 0$ such that

$$|\Phi_0(q)| \leq C_\Phi \quad \text{for all } q \in M.$$

Therefore,

$$|\Phi_0(q_T) - \Phi_0(q_0)| \leq |\Phi_0(q_T)| + |\Phi_0(q_0)| \leq 2C_\Phi.$$

After projecting by F , by Cauchy-Schwarz,

$$|F^T(\Phi_0(q_T) - \Phi_0(q_0))| \leq |F| |\Phi_0(q_T) - \Phi_0(q_0)| \leq 2|F|C_\Phi.$$

Hence

$$|R_T^F| = \left| \frac{F^T(\Phi_0(q_T) - \Phi_0(q_0))}{\sqrt{T}} \right| \leq \frac{2|F|C_\Phi}{\sqrt{T}}.$$

Thus

$$R_T^F = O(T^{-1/2}).$$

This shows that the corrector part naturally has order $T^{-1/2}$.

1.3 Part 2: Ergodic average error

Now recall the martingale part:

$$F^T M_T = \sqrt{2} \int_0^T F^T (I_d - \nabla \Phi_0(q_s)) dW_s.$$

Equivalently,

$$F^T M_T = \sqrt{2} \int_0^T (F - \nabla(F^T \Phi_0)(q_s)) \cdot dW_s.$$

Define

$$H_F(q) := F - \nabla(F^T \Phi_0)(q).$$

Then

$$F^T M_T = \sqrt{2} \int_0^T H_F(q_s) \cdot dW_s.$$

By the quadratic variation formula for stochastic integrals,

$$\langle M^F \rangle_T = 2 \int_0^T |H_F(q_s)|^2 ds.$$

Define

$$A_F(q) := 2|H_F(q)|^2.$$

Then

$$A_F(q) = 2|F - \nabla(F^T \Phi_0)(q)|^2,$$

and therefore

$$\langle M^F \rangle_T = \int_0^T A_F(q_s) ds.$$

So the finite-time variance rate is

$$\frac{1}{T} \langle M^F \rangle_T = \frac{1}{T} \int_0^T A_F(q_s) ds.$$

By the ergodic theorem, in infinite time,

$$\frac{1}{T} \int_0^T A_F(q_s) ds \longrightarrow \int_M A_F(q) \mu(dq).$$

Moreover,

$$\sigma_F^2 = F^T D F = \int_M A_F(q) \mu(dq).$$

Thus the finite-time ergodic average error is

$$E_{\text{erg}}^F(T) := \left| \frac{1}{T} \int_0^T A_F(q_s) ds - \int_M A_F(q) \mu(dq) \right|.$$

Under a spectral gap assumption, the time average has a mean-square error estimate of the form

$$\mathbb{E} \left[\left| \frac{1}{T} \int_0^T A_F(q_s) ds - \int_M A_F(q) \mu(dq) \right|^2 \right] \leq \frac{C}{T}.$$

That is,

$$\mathbb{E} [(E_{\text{erg}}^F(T))^2] \leq \frac{C}{T} = O(T^{-1}).$$

Then by Cauchy–Schwarz,

$$\mathbb{E} [E_{\text{erg}}^F(T)] \leq \sqrt{\mathbb{E} [(E_{\text{erg}}^F(T))^2]} \leq \frac{C}{\sqrt{T}}.$$

Therefore

$$\mathbb{E} [E_{\text{erg}}^F(T)] = O(T^{-1/2}).$$

So the ergodic average error also naturally contributes an order $T^{-1/2}$ term.

1.4 Part 3: Martingale CLT error

The remaining part is the martingale central limit theorem error. Formally,

$$\frac{M_T^F}{\sqrt{T}} \approx N(0, \sigma_F^2).$$

Define

$$E_{\text{mar},K}^F(T) := d_K \left(\mathcal{L} \left(\frac{M_T^F}{\sqrt{T}} \right), N(0, \sigma_F^2) \right).$$

In terms of distribution functions,

$$E_{\text{mar},K}^F(T) = \sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{M_T^F}{\sqrt{T}} \leq x \right) - \Phi_{\sigma_F}(x) \right|.$$

The qualitative martingale central limit theorem says that, if the normalized quadratic variation converges to the deterministic variance σ_F^2 , then

$$\frac{M_T^F}{\sqrt{T}} \Rightarrow N(0, \sigma_F^2).$$

Here the normalized quadratic variation is

$$\frac{1}{T} \langle M^F \rangle_T = \frac{1}{T} \int_0^T A_F(q_s) ds,$$

and Part 2 explains why this converges to

$$\int_M A_F d\mu = \sigma_F^2.$$

For a convergence rate, we would need a quantitative martingale CLT, or a Berry–Esseen type bound for martingales. The expected form is

$$E_{\text{mar},K}^F(T) \leq \frac{C_F}{\sqrt{T}} + C_F \mathbb{E} [E_{\text{erg}}^F(T)].$$

Using Part 2,

$$\mathbb{E} [E_{\text{erg}}^F(T)] \leq \frac{C}{\sqrt{T}},$$

so this suggests

$$E_{\text{mar},K}^F(T) = O(T^{-1/2}).$$

1.5 Combining the three parts

Recall

$$Y_T^F = R_T^F + \frac{M_T^F}{\sqrt{T}}.$$

The three possible sources of finite-time error are:

corrector error + ergodic average error + martingale CLT error.

From Part 1,

$$R_T^F = O(T^{-1/2}).$$

From Part 2,

$$\mathbb{E} [E_{\text{erg}}^F(T)] = O(T^{-1/2}).$$

From Part 3, assuming a quantitative martingale CLT,

$$E_{\text{mar},K}^F(T) = O(T^{-1/2}).$$

Therefore it is natural to expect

$$d_K(\mathcal{L}(Y_T^F), N(0, \sigma_F^2)) = O(T^{-1/2}).$$

Equivalently,

$$d_K\left(\mathcal{L}\left(\frac{F^T(Q_T - Q_0)}{\sqrt{T}}\right), N(0, F^T D F)\right) = O(T^{-1/2}).$$

This explains why the numerical log-log slope should be close to $-1/2$. “Finally, if a quantitative martingale CLT gives an $O(T^{-1/2})$ estimate for the martingale part, then the total finite-time distributional error satisfies

$$d\left(\mathcal{L}\left(\frac{F^T(Q_T - Q_0)}{\sqrt{T}}\right), N(0, F^T D F)\right) = O(T^{-1/2}).$$

This explains why the numerical convergence rate is expected to be close to

$$T^{-1/2}.$$